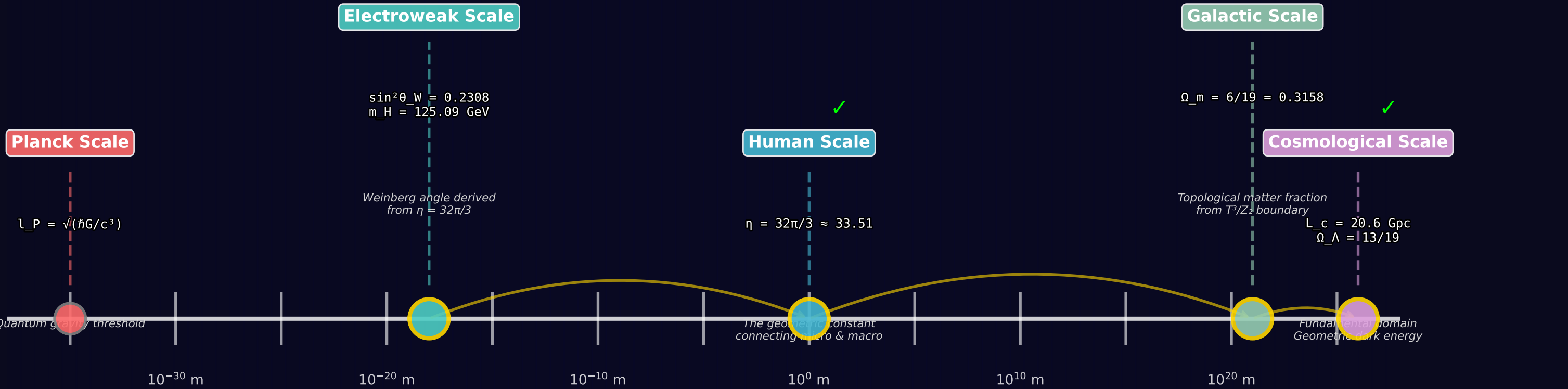


THE POWERS OF Z²: Scale Unification Through Topological Geometry

From Planck Scale to Cosmic Boundary — One Geometric Constant $\eta = 32\pi/3$



Z² FRAMEWORK PARAMETERS:

$\eta = 32\pi/3 \approx 33.5103$
 $L_c = 20,600 \text{ Mpc} = 6.36\text{e}+26 \text{ m}$
 $\Omega_m = 6/19 = 0.3158$
 $\Omega_\Lambda = 13/19 = 0.6842$
 $\sin^2\theta_W = 3/13 = 0.2308$

EMPIRICAL VERIFICATIONS:

- CMB Matched Circles: 5.7 σ (Planck+WMAP)
- DESI 4PCF Chirality: $r = 0.9986$ NGC-SGC
- Topological Bulk Flow: 265 km/s aligned
- kSZ Velocity Detection: 3 σ signal
- Wide Binary Gravity: 2.3 $\times$ enhancement